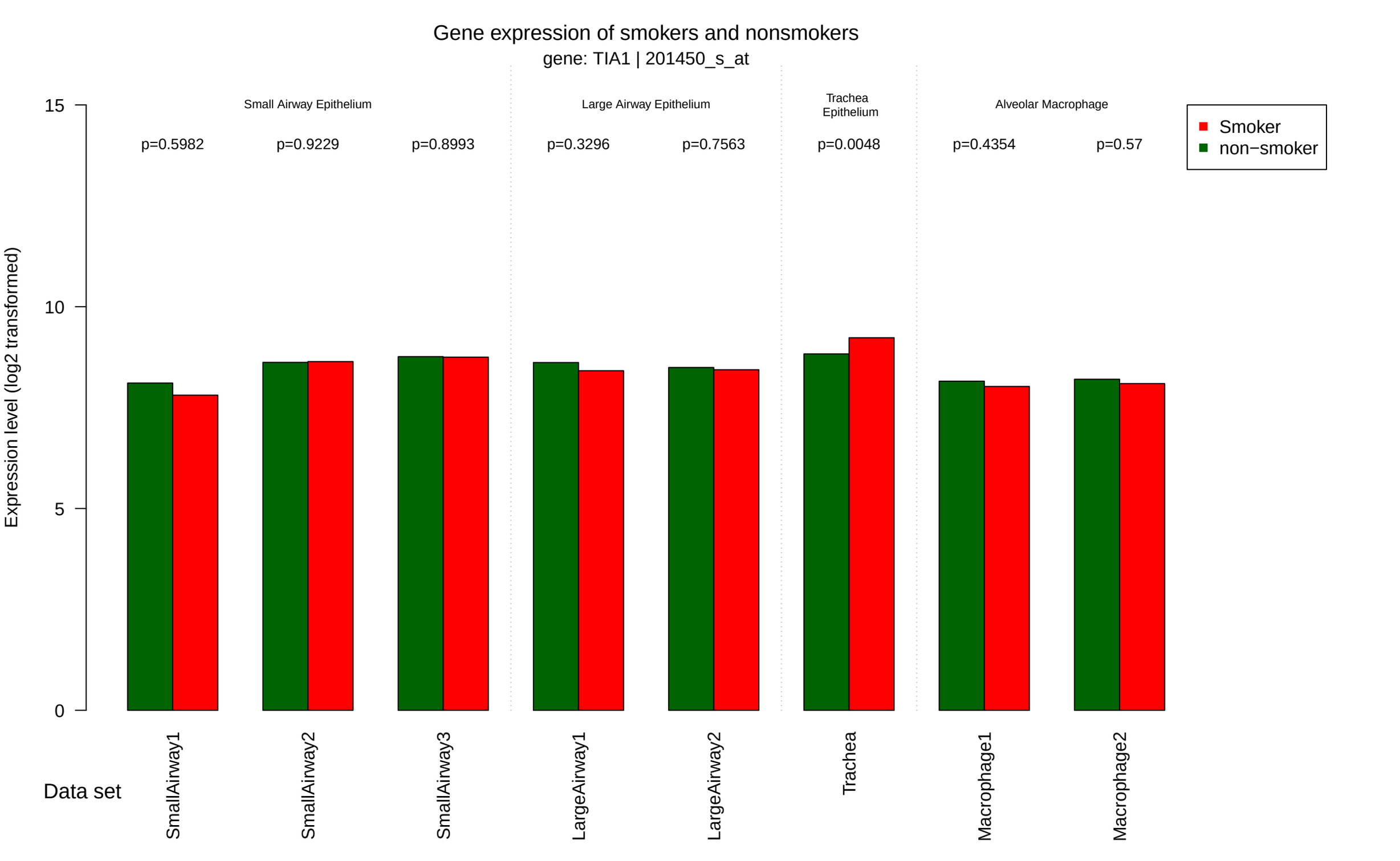
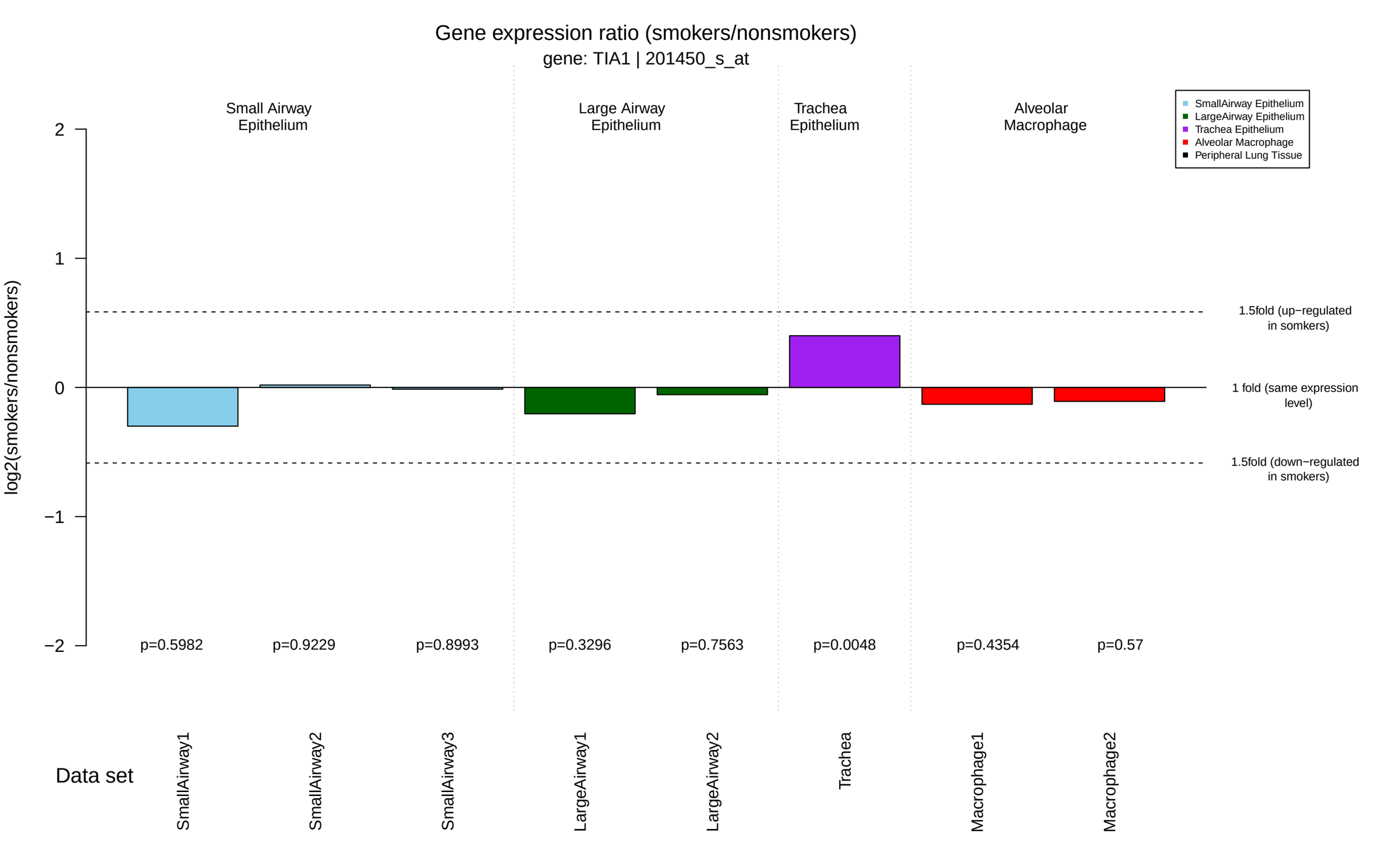
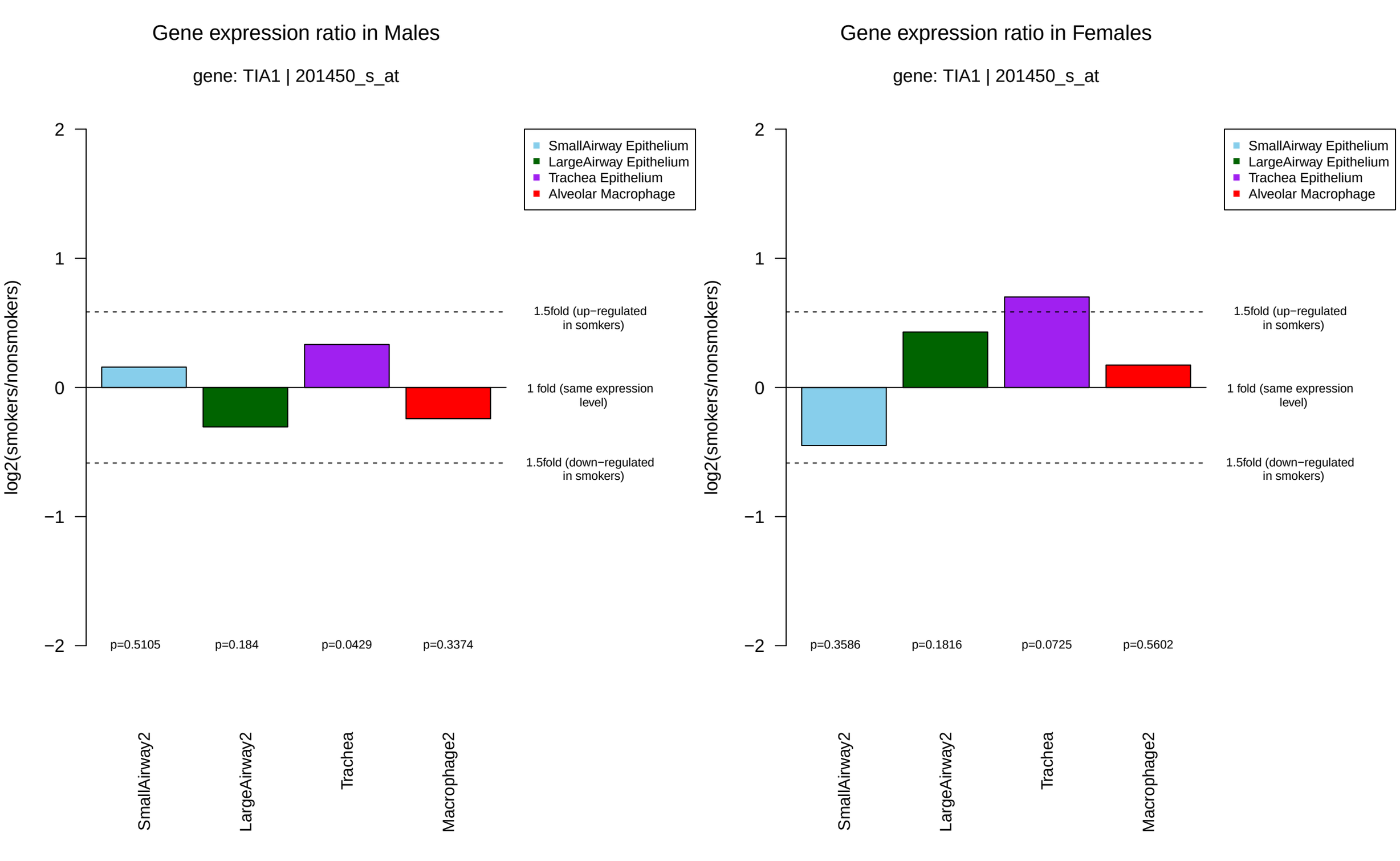
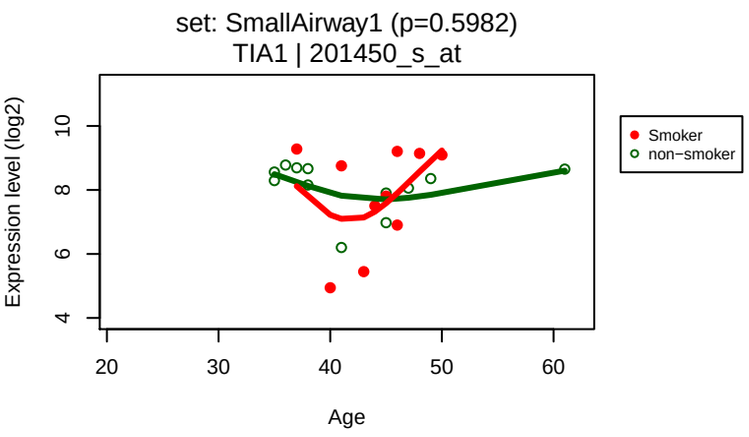
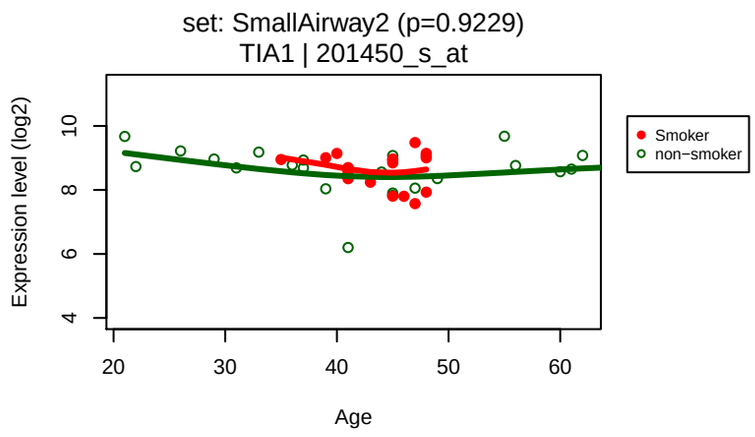
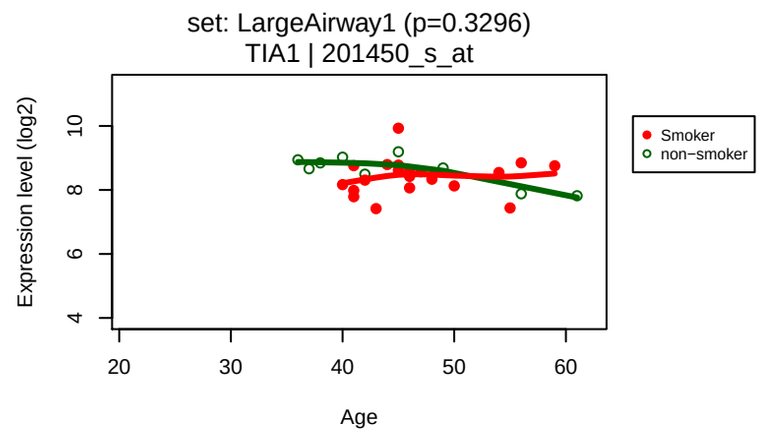

Gene expression of smokers and nonsmokers
gene: TIA1 | 201450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5982
p=0.9229
p=0.8993
p=0.3296
p=0.7563
p=0.0048
p=0.4354
p=0.57
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

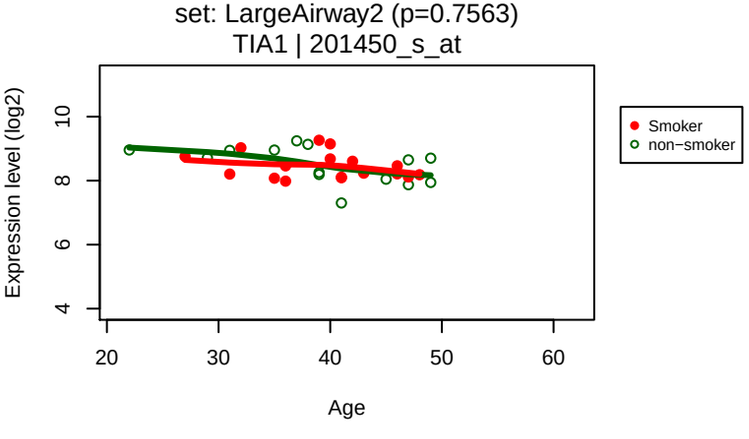

Gene expression ratio (smokers/nonsmokers)
gene: TIA1 | 201450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5982
p=0.9229
p=0.8993
p=0.3296
p=0.7563
p=0.0048
p=0.4354
p=0.57
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

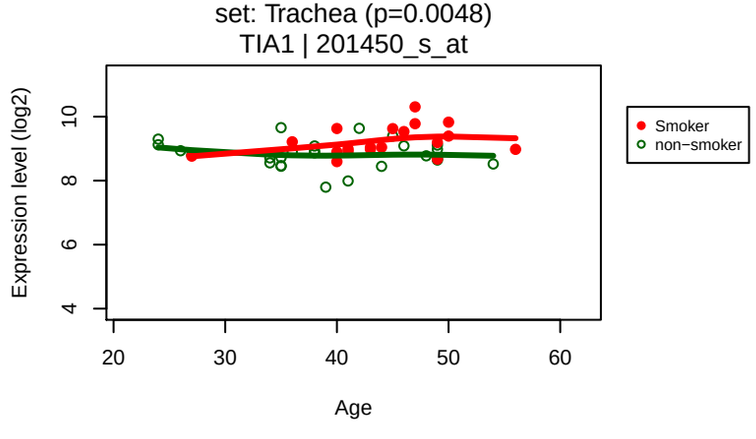

Gene expression ratio in Males
gene: TIA1 | 201450_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5105
p=0.184
p=0.0429
p=0.3374
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIA1 | 201450_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3586
p=0.1816
p=0.0725
p=0.5602
SmallAirway2
LargeAirway2
Trachea
Macrophage2

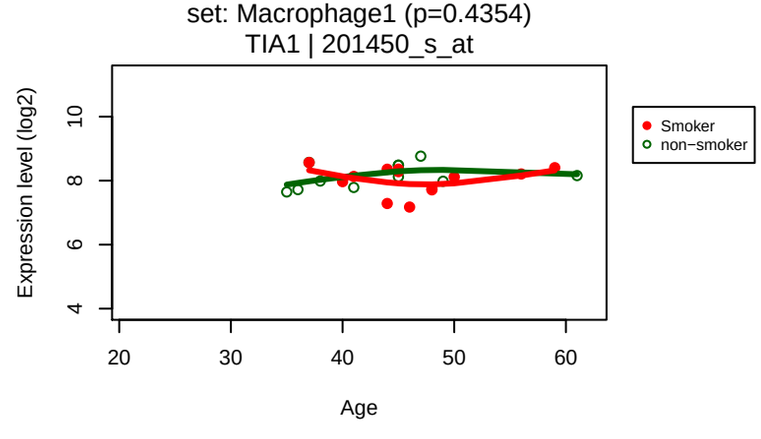

set: SmallAirway1 (p=0.5982)
TIA1 | 201450_s_at
Expression level (log2)
Age
Smoker
non−smoker

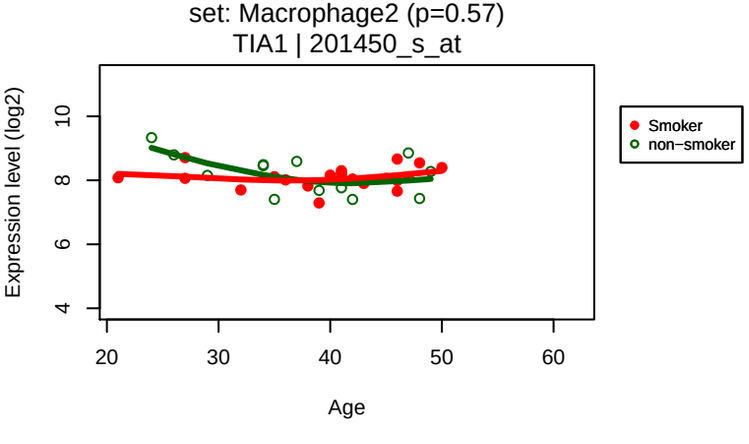

set: SmallAirway2 (p=0.9229)
TIA1 | 201450_s_at
Expression level (log2)
Age
Smoker
non−smoker

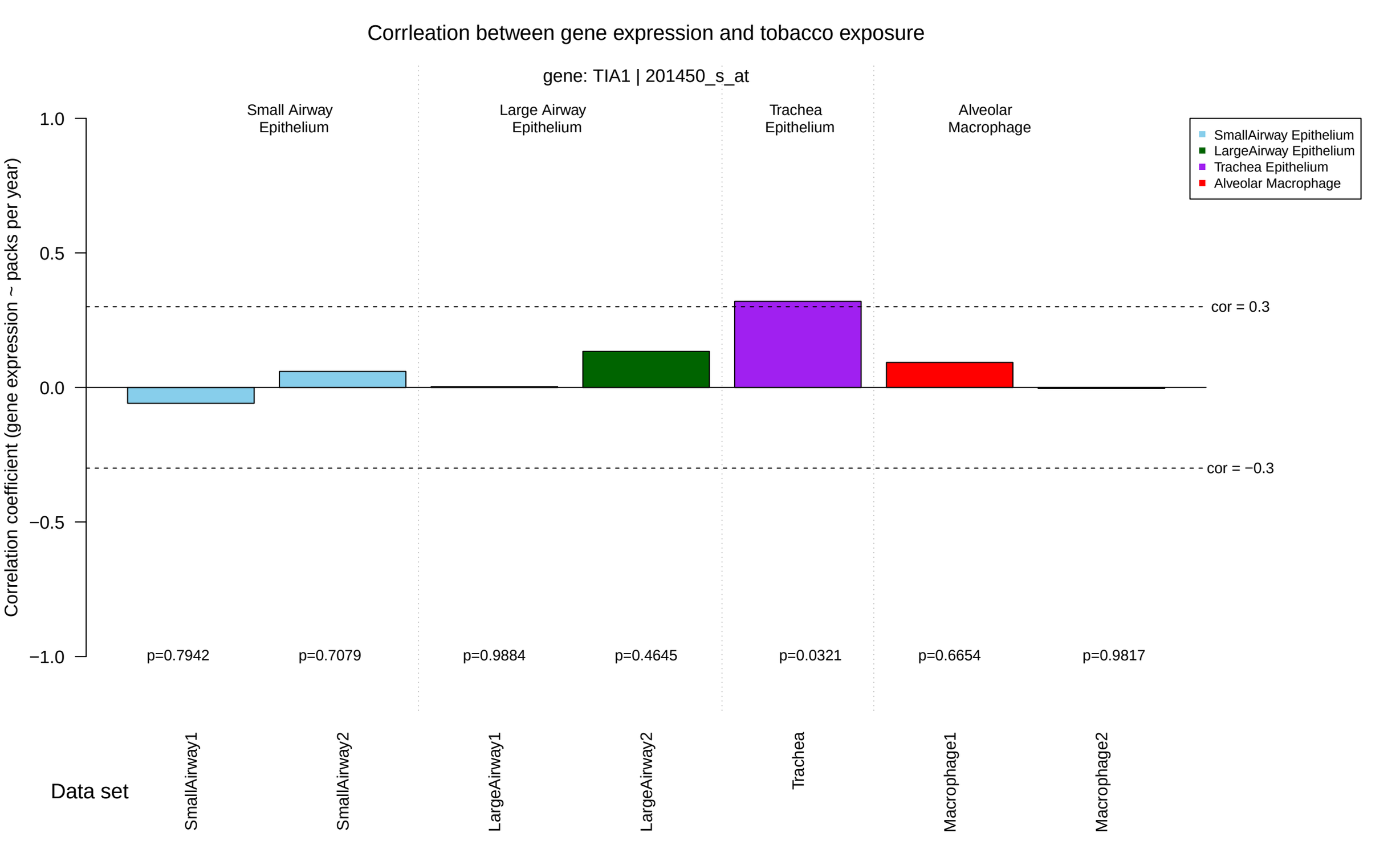

set: LargeAirway1 (p=0.3296)
TIA1 | 201450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7563)
TIA1 | 201450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0048)
TIA1 | 201450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4354)
TIA1 | 201450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.57)
TIA1 | 201450_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIA1 | 201450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7942
p=0.7079
p=0.9884
p=0.4645
p=0.0321
p=0.6654
p=0.9817
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2